



A game-theoretic analysis of traditional song selection rules in rhythm game tournaments

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ABSTRACT

In competitive rhythm gaming, the selection of songs (or “charts”) to be played is a crucial factor that can significantly influence the outcome of the match. Building on this importance, rhythm game tournaments often incorporate strategic elements into the song selection process. Instead of predefined selections, match songs are determined through player interactions in a mini-game, enhancing the tournament’s entertainment value. This selection process must balance various factors, including fairness and entertainment value, making its rule design essential for improving the overall tournament experience. Reflecting these diverse requirements, a variety of selection methods are actively used in both fan-driven and official tournaments. In this paper, we mathematically analyze the fairness of these selection processes using game theory, for the purpose of providing insights into the design of rhythm game tournament rules. Focusing on three song selection methods confirmed to be used in real tournaments, we conduct a comparative analysis with an emphasis on fairness and entertainment value.

1. Introduction

Rhythm games challenge players to perform a sequence of timed button presses synchronized with background music. The combination of button sequences and music is called a *chart*, which functions similarly to levels in other genres. A single song may have multiple charts, each varying in difficulty [1] and structure.

Rhythm games have tournaments where players compete against each other, as with other genres of games. However, the characteristics of rhythm game tournaments differ from other games in two main ways: “lack of interaction between players during gameplay” and “each players’ strengths and weaknesses about the chart for the match, which is an environment shared between players”. Rhythm games, at their core, test a player’s ability to execute precise actions in synchronization with music. Unlike other competitive game genres, where players interact with each other directly during gameplay, rhythm games are primarily tests of individual performance. However, in a competitive setting, where players face off under structured tournament formats, an additional strategic layer emerges. One of the most critical strategic elements in rhythm game tournaments is the selection of charts for each match, which we refer to as *chart selection*. In usual tournament settings, players compete for achieving a higher score on the same

chart, as shown in Fig. 1. Since different charts demand different skill sets [2], the chart that a player is most comfortable with is not necessarily the chart that can maximize the utility of the match. Therefore, players must assess their skill set, detect differences between the opponent, and choose the best chart that maximizes their utility in order to gain a significant competitive edge. This adds a *metagame* layer to the tournament experience, requiring players not only to master execution but also to make strategic decisions before the match even begins.

Given the significance of chart selection, many tournaments introduce a structured decision-making process – often through a predefined minigame – to determine the chart used in a match. A notable example of the importance of chart selection phase can be seen in the “BEMANI PRO LEAGUE¹”, an official esports league for rhythm games run by Konami Amusement. Before the actual performance, players are given time to decide on their chart choices. This phase is named as the “STRATEGY TIME”. The presence of this dedicated time window, along with its naming, suggests that rhythm game developers recognize the strategic depth involved in chart selection and have integrated it as a key element in tournament formats.

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¹ <https://p.eagate.573.jp/game/bpl/> <Retrieved on Jan. 22, 2025>.

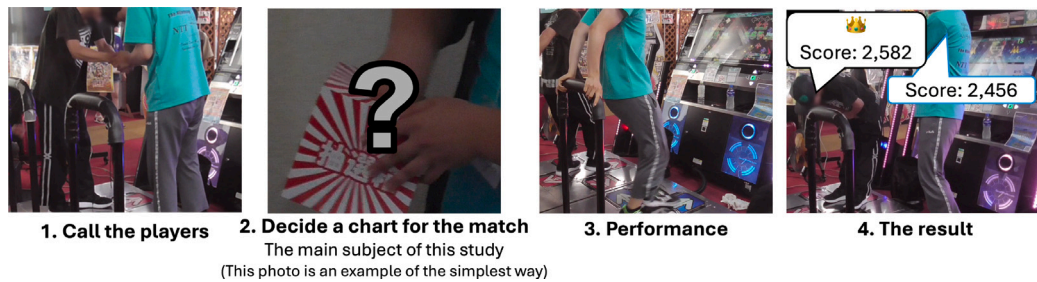


Fig. 1. An example of one match in a rhythm game community tournament. (The pictures are captured from the tournament footage organized by the second author.).

1.1. The challenge of tournament rule design

One of the core responsibilities of tournament organizers – whether for official or community tournaments – is designing the tournament rules. A well-designed rule set ensures fair play, engages both players and spectators, and accommodates logistical constraints. Since most games are not restricted to a singular way of enjoying them, these rules help on highlighting the various ways the games can be enjoyed.

Chart selection phases introduce an additional layer of interaction between players, which is usually difficult to design in rhythm games due to their individual-focused nature. As the tournament gathers many players of the game, communicating with others can be a reason to participate. Therefore, making the interaction between the players, both for the spectators and the players themselves, is also an important aspect of rhythm game tournament design.

The important considerations for designing rhythm game tournaments can be summarized as follows:

- *Fairness*: Ensuring that no player has an undue advantage due to chart familiarity.
- *Entertainment*: Maintaining excitement for both players and spectators.
- *Practical Constraints*: Adapting to tournament durations and venue limitations.

1.2. The evolution of rhythm game tournaments and community influence

Because a player's familiarity with a chart can significantly impact performance, tournaments have employed diverse methods for determining which charts are played, each with different implications for fairness and strategy. While some methods have shifted over time, others have remained in use for decades, standing the test of time.

Rhythm game tournaments exist in both official and grassroots settings, evolving over decades of time. The number and scale of these tournaments have grown in recent years with the increasing popularity of “esports”. For example, “BEMANI PRO LEAGUE” is the official esports league of three rhythm games developed by Konami Amusement: *Beatmania IIDX*, *SOUND VOLTEX*, and *Dance Dance Revolution (DDR)*. *Taiko no Tatsujin*² by Bandai Namco Entertainment Inc., *maimai*³ by SEGA, and *HATSUNE MIKU: COLORFUL STAGE!*⁴ by SEGA and Colorful Palette also have official international championships. With the growing competitive scenes in rhythm games, some titles have officially implemented the player-versus-player (PvP) mode, accessible to every user. “ARENA MODE” in *Beatmania IIDX* and “LADDER MATCH” in *DJMAX RESPECT V*⁵ developed by Rockey Studio are examples of PvP modes.

Along with these official tournaments, *community tournaments*, often managed by passionate volunteers, play a crucial role in shaping tournament culture. Some of the earliest DDR tournaments date back to at least 1999.⁶ In this way, the culture of rhythm game tournaments has developed both officially and driven by the gaming community. Many chart selection methods used today have originated and evolved through these informal community tournaments.

Playing games is one of the most popular forms of entertainment. While playing in individual homes was the mainstream with Nintendo Entertainment System and PlayStation, playing in public spaces has also been popular for years: amusement arcades and game centers. There are various types of games in such public spaces, which have provided not only enjoyable games but also a fan community. A fan community is essential to growing and maintaining the culture of games. The community's activities are varied [3], including discussing [4] and coaching [5] the strategies, livestreaming the gameplay [6], making fan fiction [7], and even helping developers through playtesting [8,9] or reviewing [10]. Such public space and the community naturally bring ideas for new styles of enjoying games to players, which is an example of positive impacts of fan communities to the games itself.

Providing an in-depth analysis of various chart selection rules is valuable from the perspective of the players as well. There are some popular rules for the chart-deciding process that have stood for a long time. These rules have been spread informally by the chain of imitation; an organizer borrows the rule from the past tournament in which the organizer participated as a player. Enthusiastic players, like the second author, have sensed the strategy and interest of the rule. However, despite the popularity of the rule in the community of rhythm games, the qualities of these rules have not been studied in academic format as far as we know. It is thus important not only for the tournament organizers, but also for the players, to provide an in-depth mathematical analysis of the chart selection methods.

While competitive rhythm gaming emphasizes individual performance, tournaments serve as more than just skill-based competitions — they also function as social events that connect players and foster a sense of community. This aspect is especially crucial for engaging newcomers and sustaining interest in the scene. Therefore, offering valuable insights into tournament design plays a key role in cultivating and expanding the rhythm gaming community.

1.3. Related works

In competitive gaming, *metagames* refer to strategic decision-making beyond in-game execution. Metagames exist in all competitive environments where players must make decisions beyond immediate gameplay mechanics. Some existing papers address the metagame. Shen et al.

² <https://taiko.namco-ch.net/taiko/en/wcs2024/> <Retrieved on Jan. 30, 2025>.

³ <https://maimai.sega.jp/kop5th/> <Retrieved on Feb. 3, 2025>.

⁴ <https://pjsekai.sega.jp/worldchampionship2024spring/en/index.html> <Retrieved on Jan. 30, 2025>.

⁵ https://store.steampowered.com/app/960170/DJMAX_RESPECT_V/ <Retrieved on Jan. 19, 2025>.

⁶ We confirmed that there was a DDR community tournament in 1999. Although we cannot find the original information from those days, we found the statement about the tournament on the tournament's website held in 2003. <https://www.gamersnest.com/rca/pssp2003/about.html> <Retrieved on Jan. 22, 2025>.

studied the strategy in the metagame of the Multiplayer Online Battle Arena (MOBA) game [11]. Reis et al. presented a framework of competition about the AI that balances the metagame of Pokémon battles [12].

There are some studies about the design of tournaments, not only for esports but also for physical sports and mind sports. One of the mainstream of tournament design studies is matching the participants, including creating tournament brackets [13] and estimating the player's skill [14].

The chart decision process in the rhythm game tournament is an attempt to reach an agreement between the players about the shared environment of actual gameplay. In this aspect, the process is related to a consensus building [15].

By analyzing rhythm game chart selection through a game-theoretic lens, we can gain insights into how different selection methods affect strategic depth, fairness, and competitive balance. Additionally, understanding these mechanisms may inform broader tournament design strategies across various gaming genres.

1.4. Scope and contributions of this study

Despite the long-standing presence of chart selection rules in rhythm game tournaments, to the best of the knowledge of the authors, there has been no formal academic study of its strategic implications. This paper bridges that gap by analyzing how different chart selection rules impact fairness, entertainment value, and competitive balance.

Specifically, our contributions are as follows:

- We document and formalize rhythm game tournament rule designs that have traditionally been shared informally within gaming communities.
- We compared the rules unique to rhythm game tournaments with a rule also found in other genres. Through this, we examined the characteristics of rhythm game tournaments and the suitable tournament scenes for each rule.
- We analyze the strategic characteristics and entertainment value of different rule designs using game theory.

By providing a structured mathematical analysis of these selection methods, this study aims to contribute not only to the academic understanding of game theory in esports but also to the gaming community by offering insights for tournament organizers seeking to create engaging and fair competitive environments.

1.5. Structure of this paper

The remainder of this paper is organized as follows. Section 2 describes the prerequisites of the paper, explaining the details of rhythm game dynamics that are at the focus of our analysis. Section 3 describes the details of chart selection methods that are used in actual tournaments, which are the main subjects of the analysis in the present paper. Sections 4–6 provide a game-theoretic analysis of the three chart selection methods described in the previous section. Section 7 analyzes specific examples to compare the characteristics of each rule. Section 8 discusses the implications of our findings, including their potential application to other game genres. Finally, Section 9 summarizes our contributions and outlines future research directions.

2. Prerequisites of rhythm game competitions

Here we will summarize the prerequisites of our analysis. In this section, we provide a detailed explanation of the specifics of rhythm game competitions that are important for our discussion, followed with a brief summary of the focus of our analysis.

2.1. Rhythm games and the necessity of chart selection

In casual settings, the players can freely choose the charts they want to play based on their skill or mood, since there are no time constraints. Most titles of rhythm games have a rich variety of songs, each usually equipped with multiple charts. For example, DDR includes over 1200 songs and approximately 5000 charts as of Jan. 19, 2025.⁷

The player's performance is evaluated with an in-game score. The accurate actions with the chart results in a higher score. To achieve a higher score on a specific chart, the player should capture the characteristics of the chart [2] through repetitive practice, in addition to establishing the base skill of the game by playing many charts.

In competitive settings, chart selection becomes an important strategic element of the match. Due to resource constraints, it is usual for only one chart to be used in a match, especially in early rounds of the tournament. When the time bank of the tournament is large, each match can simply consist of multiple charts, somewhat resolving the conflict arising from the necessity of choosing one chart. However, when there are many participants, there may not be enough time to play multiple charts. Such cases usually arise in early stages of the tournament. Therefore, there are actual cases when only one chart is used for a match, justifying the setting of our analysis.

In some games, the players' physical energy is another practical resource constraint. This is mainly a problem if the game demands extreme physical energy such as DDR. In these cases, using more charts in a match demand the players to expend their stamina, which would likely lower their performance in later charts or rounds. Limiting the match to one chart important in such cases as well.

Due to the strategic importance of chart selection, generally, it is thus the tournament organizer's responsibility to design rules that treat all of the players in the match fairly. The design of these rules can significantly impact the dynamics and the experience of the tournament.

2.2. An illustrative counterexample of unfair chart selection rules

Here, we illustrate the importance of fairness of chart selection rules by showing a counterexample of an unfair rule design.

Consider an unfair chart selection rule which we shown in Fig. 2. In this rule, two players must choose one chart from a chart pool of three, but only one of the players have the right to choose a chart. This rule is unfair in the following aspects:

- Only one player has an opportunity for intervention in the process.
- The skills of the non-intervening player is ignored.
- If the chart pool is known before the tournament, the intervening player can practice a specific chart in advance, while hiding their selection to the other player, placing a disadvantage on the non-intervening player.

To win the match, the non-intervening player must have an outstanding skill, enough to surpass the opponent's score even with the opponent's best preference. Therefore, the right to select the chart provides the intervening player a significant advantage, even though it is outside the actual gameplay mechanics.

2.3. Chart selection methods in practice

One of the ways to ensure fairness in the chart selection process is by introducing interactions between two players in the match. This interaction is essentially a *minigame* played by the contestants, in order to choose a chart.

A typical rhythm game match in a tournament with a chart selection phase is shown in Fig. 1. Here, the procedure for the match is as follows:

⁷ <https://p.eagate.573.jp/game/ddr/ddrworld/music/index.html?offset=0&filter=7>

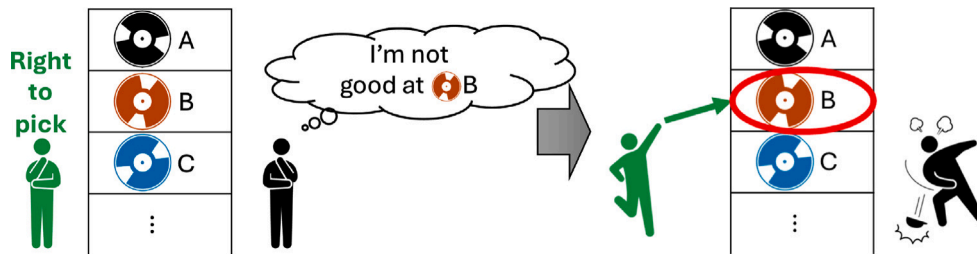


Fig. 2. An example of an unfair rule.

1. The tournament committee calls the players up to the stage.
2. Each match features two players competing on the same chart. Only one chart is used per match. The chart is decided based on player interactions, following a predefined chart selection rule. Rules are later described in Sections 3.2–3.4.
3. Each player plays the decided chart. The player with the best in-game score is the winner of the match.

The “candidate chart pool” in Step 2 is a set of charts chosen by the committee, and could generally have any number of charts. The process of creating the chart pool varies by each tournament. In some cases, the tournament organizers preselect the charts based on their preferences, while in others, the charts are randomly drawn from a larger pool on the day of the event.

2.4. Benefits of introducing interactive chart selection methods

Introducing such minigames emphasizes the tournament’s nature of the game as a PvP battle. As mentioned in the introduction, in rhythm games, PvP mechanics are more challenging to highlight in tournaments compared to other game genres. This different from other genres such as fighting games or MOBA, where the mechanics are designed for playing against other players, ensuring sufficient interaction between players in the actual gameplay. To design rhythm game tournaments as a PvP competition, they should contain interaction between players other than the actual gameplay. In today’s tournaments, using meta minigames with interaction in the chart selection process has become the primary trend.

For the audience, the minigame could be one of the attractions of the tournament. Rhythm game tournaments are not only for competition but also for communication among people who share the same interests. The players’ behavior in the minigame is a good topic for communication, especially for post-match discussions about the player’s strategies.

3. Overview of chart selection rules

In this paper, we analyze and discuss three kinds of chart selection rules: *Mind Game*,⁸ *Sequential Veto*, and *Simultaneous Veto*. As mentioned in Section 1.2, there have been numerous tournaments with various rulesets. This makes it impossible to cover all rules comprehensively. Therefore, this paper analyzes these three rules, which we confirmed to have been used in actual rhythm game tournaments or the official PvP competitive mode, as traditional examples.

Mind Game is a rule that has been used in DDR tournaments held in Japan for over 15 years.⁹ *Sequential Veto* is primarily used in the DDR

tournament held in North America.¹⁰ *Simultaneous Veto* is based on the rule used in the official competition mode named “LADDER MATCH” of the Korean rhythm-based game “DJMAX RESPECT V¹¹”.

We confirmed that *Sequential Veto* is used in some game tournaments other than rhythm games with some tweaks.¹² We also found early research about a similar example in the field of economics [16]. On the other hand, to the best of our investigation, the remaining two rules seem to be used only in rhythm game tournaments. We, therefore, expect that our findings will imply the unique aspects of rhythm game tournaments and be helpful for the active community of the players.

In this section, we first explain the focus of our analysis, followed by a detailed explanation of each chart selection rule.

3.1. The focus of our analysis

The purpose of the present paper is to analyze the properties of existing rules. We are interested in properties such as fairness and the quality of entertainment. For the purpose of comparing multiple methods, we fix the number of players, chart pool size, and the number of charts to be used in a single match, and focus on cases where two players decide on one chart from a pool of three charts. Generalizing the size of the chart pool is discussed in Section 8.3.2.

Additionally, we optimize the result of a single match, and do not consider the interactions between multiple matches. As an example of a situation that is not covered, some players may not need to win specific matches to save their energy for later matches in a round-robin tournament. This simplification is justified since such a situation is not likely to occur in a single-elimination tournament where one loss disqualifies the player.

3.2. Mind game

The Mind Game rule proceeds as follows, as shown in Fig. 3:

1. Each player checks the candidate chart pool and then secretly chooses one chart from the pool.
2. When both players are ready, the committee makes a cue, and both players reveal their choices simultaneously.
 - (a) If both players select the same chart, the chart is decided for the match.

⁸ Mind Game is a English translation of Japanese term: 心理戦 (Shinri-sen).

⁹ As far as we have confirmed, this rule was used in the tournament held in Japan in 2006 (<https://www.gamersnest.com/zyx/score.html> <Retrieved on Jan. 19, 2025,>) and 2024 (<https://butoutaikaisou.wixsite.com/doublestronger/blank-1> <Retrieved on Jan. 19, 2025,>) Note that the former tournament employs a variant of the Mind Game rule that chooses two charts from a pool of three.

¹⁰ For example, a tournament held in North America in 2017 (https://docs.google.com/document/d/1Yn92EJY0X14ihV_3nhmb65ZRAGADMZY7IPoi0vZfCs/edit?usp=sharing <Retrieved on Jan. 19, 2025,>) uses the Sequential Veto rule. The number of charts in the match varies depending on the round.

¹¹ In this mode, five charts are presented as candidates for a match, and three are used.

¹² We found examples in the tournament of first-person shooter game VALORANT by Riot Games (<https://cdn.sanity.io/files/dsfx7636/news/502599ce1abe438ecbe8139f87df2e10b048ad98.pdf> <Retrieved on Apr. 28, 2025,>) and the fighting game Super Smash Bros. by Nintendo (https://docs.google.com/document/d/e/2PACX-1vS2EvECU9nErMyHgd431PgcfT1Is_VY0JQRJyqY90-MDS0G3SYOfXAMvrFLhYmItLYo8vAot2x0Bw2l/pub <Retrieved on Apr. 28, 2025,>).

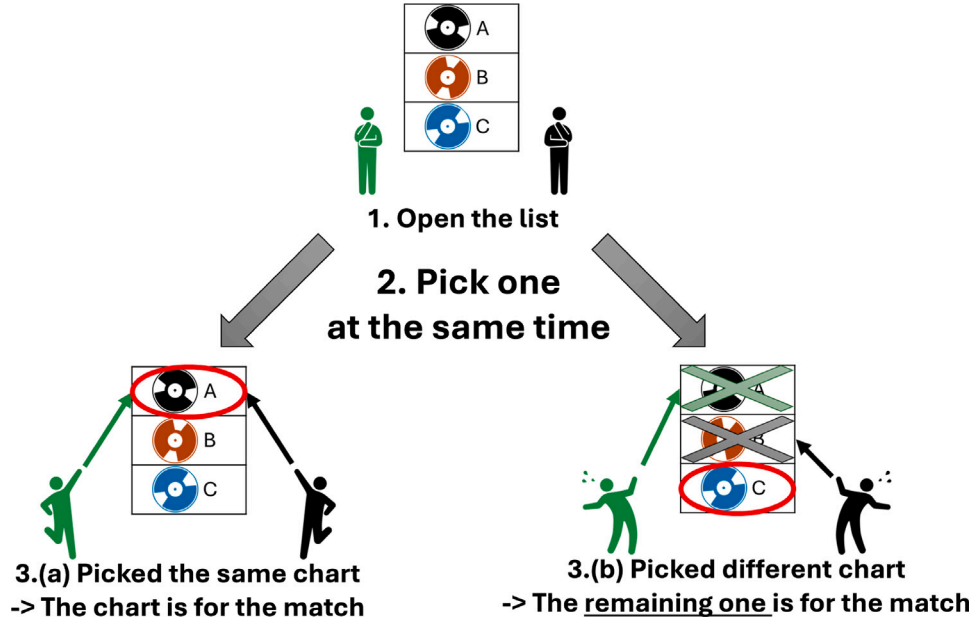


Fig. 3. The flow of the Mind Game rule.

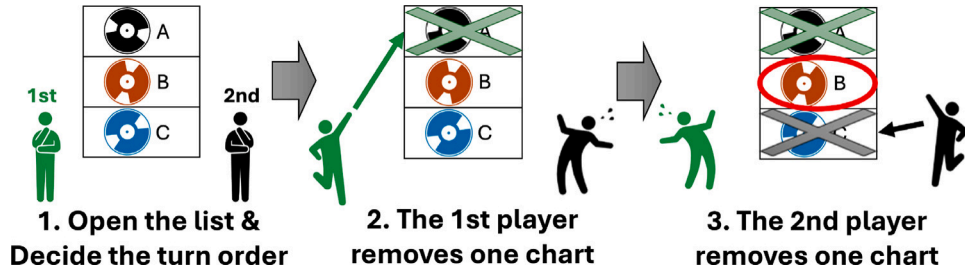


Fig. 4. The flow of the Sequential Veto rule.

- (b) If each player selects a different chart, the *remaining chart* that neither chooses is decided for the match.

The outcome of Mind Game is hard to expect since the player's choice may be accepted or rejected depending on the opponent's choice.

3.3. Sequential veto

The Sequential Veto rule works in the following procedure, as shown in Fig. 4:

1. Decide the order of action for each player. The candidate chart pool is then shown for both players.
2. The first player vetoes one unwanted chart from the pool.
3. The second player vetoes one from the remaining two charts in the pool.
4. The remaining chart in the chart pool is used for the match.

Sequential Veto is the only one of three rules in this paper where players act in turns, instead of acting simultaneously.

3.4. Simultaneous veto

The Simultaneous Veto rule works in the following procedure, as shown in Fig. 5:

1. Each player checks the candidate chart pool and then secretly chooses one chart to eliminate from it.
2. When both players are ready, the committee makes a cue, and both players reveal the veto choice at the same time.

3. The charts chosen by each player are removed from the chart pool.
4. If only one chart is remaining in the pool, that chart is used for the match. If multiple charts are remaining, the committee randomly picks one for the match from the pool.

Simultaneous Veto is similar to the Mind Game rule in that both players make their own choices simultaneously. It is also similar to Sequential Veto since the players' choices are removed from the chart pool in both rules.

4. Analysis of the mind game rule

In this section, we analyze the rules of the Mind Game rule.

4.1. Problem formulation

We formulate the Mind Game rule as a strategic game where two players simultaneously choose one of three strategies. Let the three charts in the candidate chart pool be A, B, and C, and let the participants be referred to as Player 1 and Player 2.

Following the rules described in Section 3.2, one chart is chosen for each match, depending on the players' strategies. Since each player performs differently for each chart, we assume that the payoff for each player depends on the chart chosen for the match. The payoffs for Player 1 when charts A, B, and C are chosen are denoted as a_1 , b_1 , and c_1 , respectively, and a_2 , b_2 , and c_2 for Player 2, respectively. Each player has three strategies: "declare chart A", "declare chart B", and

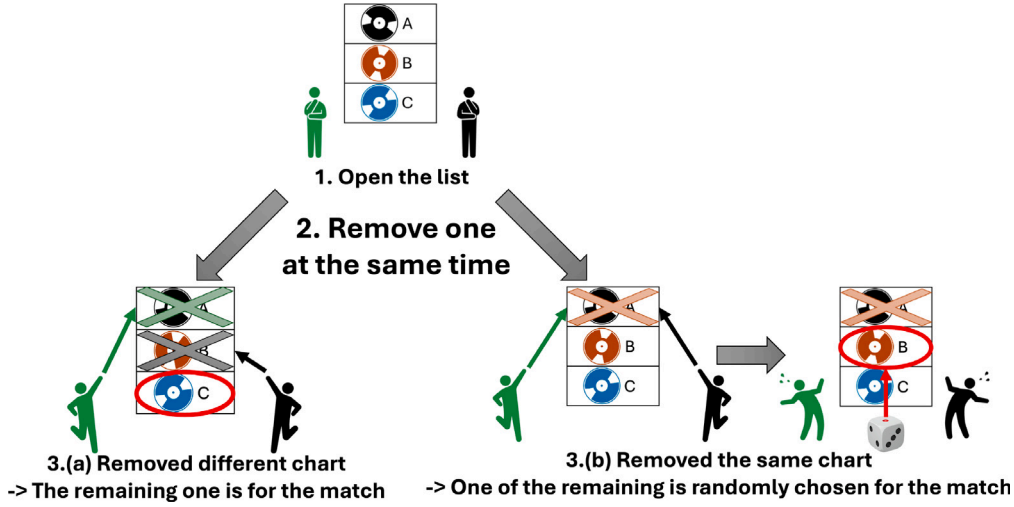


Fig. 5. The flow of the Simultaneous Veto rule.

Table 1
Payoff matrix for Players 1 and 2 in the Mind Game rule.

		Player 2		
		A	B	C
Player 1	A	a_1, a_2	c_1, c_2	b_1, b_2
	B	c_1, c_2	b_1, b_2	a_1, a_2
	C	b_1, b_2	a_1, a_2	c_1, c_2

“declare chart C, referred to as A, B, and C, respectively. Summarizing these conditions, the Mind Game rule can be formulated as a strategic game with a payoff matrix as shown in Table 1.

4.2. Modeling payoffs

To model the Mind Game rule, the remaining task is to define the values of the payoffs $a_1, b_1, c_1, a_2, b_2, c_2$. In the most straightforward setting, these payoffs are the winning probabilities for each player, which depend on the relative proficiency of each player for each chart. If we ignore ties, this will always produce a zero-sum game, since the sum of the winning probabilities always equals 1, regardless of the proficiency of the players.

However, there are various scenarios when such zero-sum payoffs are not necessarily appropriate. Altering the payoff designs significantly affects the properties of the game. Here, we will discuss various possibilities of the payoff designs.

4.2.1. Zero-sum setting

As mentioned before, a straightforward payoff design is to use expected win rates as the payoff values. For instance, consider the case where the precise skill levels of the players for each chart are common knowledge, and thus the win rates for each chart are common knowledge as well. When we ignore the possibility of ties, since the sum of the expected win rates is always 1, this method of setting payoffs results in a zero-sum game.

4.2.2. Non-zero-sum setting

While zero-sum settings represent the most basic case, non-zero-sum payoff settings can also be considered. We focus on the fact that the assumption that each player's skill level is common knowledge does not necessarily hold. For example, consider a scenario where each player knows their own skill level precisely but has limited information about the opponent's skill level. In this case, each player must use their estimated values of the opponent's skill level instead of the true values

to calculate their expected win rates. When such estimated values are used as payoffs, the sum of the players' expected win rates do not necessarily be equal to one, making this a non-zero-sum game.

Other considerations for non-zero-sum payoffs might include factors such as the players' satisfaction with the chosen chart. For example, in an extreme scenario where two charts are significantly difficult for both players, choosing these charts may make the outcome of the match depend purely on luck. Some players, especially those in recreational tournaments, might consider such matches as undesirable, and derive higher utility from a match that reflects their skills. In such cases, it may be possible that both players have a unique common chart with the highest utility. Notice that such cases are impossible for non-zero-sum settings, since in zero-sum settings, the only way for the chart with the maximum utility to be common between two players, is for the utility of all of the charts to be equal for both players.

4.3. Deriving Nash equilibria

Without loss of generality, we can assume $a_1 \leq b_1 \leq c_1$ for Player 1's payoffs. For simplicity, we also assume that the payoffs a_1, b_1, c_1 are distinct, so $a_1 < b_1 < c_1$. We also assume that Player 2's payoffs a_2, b_2, c_2 are distinct as well, but do not assume the order of the sizes for Player 2. Furthermore, we consider $0 \leq a_1, b_1, c_1, a_2, b_2, c_2 \leq 1$.

Let the probabilities of Player 1 and Player 2 choosing charts A, B, and C be denoted by (p_A, p_B, p_C) and (q_A, q_B, q_C) , respectively. In our two-player matrix game with, all Nash equilibria can be derived using the support enumeration algorithm [17]. Detailed conditions and computations are provided in Appendix. Here, we enumerate all Nash equilibria obtained from the analysis and discuss their properties.

The Nash equilibria of this game depend on the relative ordering of the payoffs for Player 2 (since we have already assumed the order of Player 1's payoffs). The following sections enumerate the Nash equilibria of the Mind Game rule based on the payoff orderings.

4.3.1. When both players prefer the same chart

This corresponds to the case where the chart that maximizes each players' payoffs is the same for both players. Note that this scenario does not occur in the zero-sum game settings, since we assume that the payoffs for each chart is distinct for each player. Therefore, this case can only arise in non-zero-sum games.

Under this condition, there are seven Nash equilibria: three pure-strategy equilibria and four mixed-strategy equilibria. The three pure-strategy equilibria involve both players always choosing chart C, with each equilibrium corresponding to the players declaring strategies (A, B), (B, A), and (C, C), respectively, with probability 1.

Among the four mixed-strategy equilibria, three involve both players mixing between two charts, while the fourth involves both players declaring all three charts with equal probability.

Focusing on the expected payoffs for these equilibria, the pure-strategy equilibrium where chart C is always chosen yields the highest expected payoff. This result stems from the fact that the expected payoff of a mixed strategy is a convex combination of the payoffs for the individual charts.

4.3.2. When players prefer different charts

In this case, there is only one Nash equilibrium, where both players declare each chart with equal probability: $(p_A, p_B, p_C) = (q_A, q_B, q_C) = (1/3, 1/3, 1/3)$. A notable point here is that these probabilities do not depend on the specific values of the payoffs $a_1, b_1, c_1, a_2, b_2, c_2$.

In the case of a zero-sum game, the payoffs satisfy $a_2 = 1 - a_1$, $b_2 = 1 - b_1$, and $c_2 = 1 - c_1$. This ensures that $a_2 > b_2 > c_2$, so the chart most preferred by both players never aligns. As a result, the optimal strategy in a zero-sum game is to select each chart with equal probability, regardless of the opponent's skill or preferences.

4.4. Equilibrium selection

When multiple Nash equilibria are present, the question arises as to which equilibrium the players should follow, a problem referred to as equilibrium selection. Several criteria for equilibrium selection have been proposed, primarily categorized as Pareto dominance and risk dominance. Under Pareto dominance, the equilibrium that maximizes the total payoffs of both players is typically selected. In this paper, we assume that when multiple Nash equilibria exist, the players agree to choose the equilibrium that maximizes their combined payoffs.

In the Mind Game rule, multiple Nash equilibria occur only when both players share the same most preferred chart. In such cases, the equilibrium yielding the highest expected payoff always involves both players choosing chart C with a probability of 1. Consequently, based on the principle of equilibrium selection, the players preemptively agree to declare chart C , ensuring maximum payoffs for both. This leads to deterministic decision-making by the players in such cases.

4.5. Optimal strategies of the mind game rule

The analysis above shows that the game unfolds in two distinct ways, depending on whether both players' most preferred charts align. Therefore, when playing in the Mind Game rule, players should first assess whether their most preferred charts align. If they do, the players should cooperate to ensure the most preferred chart is chosen, in order to maximize their payoffs. If they do not, both players should declare all charts with equal probability, regardless of their skill levels or preferences.

The equilibrium in the situation where players prefer different charts is somewhat counterintuitive. This case resembles Rock–Paper–Scissors, a simple minigame with two players, three choices, and one action. The Nash equilibrium of Rock–Paper–Scissors is to pick all choices with equal probability. However, it is known that some people do not follow the Nash equilibrium when they get information about the opponent [18]. In Mind Game, the choices have meaning in the form of a chart, and there is room to imagine the bias of the opponent's pick with knowledge such as past tournament results. This makes it more difficult to understand the Nash equilibrium in Mind Game than Rock–Paper–Scissors intuitively. This counterintuitive nature is one of the interesting aspects of Mind Game.

While this paper assumes that the payoffs are common knowledge between the players, exploring extensions to settings with incomplete information would be an interesting future work. For example, players could use bluffs to influence their opponent's perception of their skill, thereby altering the opponent's estimated win probabilities and payoffs.

Building on the results of this paper, in scenarios where players can manipulate their opponent's payoffs, they might try to create the impression that their most preferred chart aligns with their opponent's. If successful, this could eliminate the probabilistic elements of the opponent's strategy. In such cases, both players could exploit this dynamic to steer the choice toward their preferred chart.

5. Analysis of sequential veto

In this section, we analyze the rules of the Sequential Veto rule, as described in Section 3.3. In this rule, players sequentially vote for a chart to be removed from a given chart pool, and the remaining chart is used for the match. There are extensive prior researches on similar methods that involve voting for eliminating candidates [19], with early works such as [16]. Games that generalize the number of participants and candidates are also studied in the context of sequential elimination [20,21].

This paper focuses specifically on music game tournaments, examining the game dynamics that arise when Sequential Veto is adopted as the chart selection rule. Accordingly, we discuss and analyze the game properties for the case of two players and three chart choices in music game tournaments.

5.1. Problem formulation

In the Sequential Veto rule, players take turns making decisions, therefore falling under the category of a sequential game. In sequential games, the solution concept typically employed is the subgame perfect equilibrium, an extension of Nash equilibrium. In this paper, we assume that players follow the subgame perfect equilibrium of the Sequential Veto rule. We use the same settings and variables for the payoff function as in the Mind Game case in Section 4.2.

5.2. Analysis

The subgame perfect equilibrium of this game depends solely on the order of the payoffs and not on their actual values. Since we can assume $(a_1, b_1, c_1) = (1, 2, 3)$ for Player 1 without loss of generality, what remains is choosing the payoffs for Player 2. Using the fact that only the order of the payoffs matter, we assume that a_2, b_2 , and c_2 are permutations of 1, 2, and 3. Notice that the zero-sum case occurs only when $(a_2, b_2, c_2) = (3, 2, 1)$, while all other cases correspond to non-zero-sum games.

Once the payoffs are determined, there are no probabilistic elements in its progression. A specific chart is always chosen with probability 1, based on the payoffs.

5.3. Special equilibrium points

This game exhibits interesting properties when the opponent's payoffs are $(a_2, b_2, c_2) = (1, 3, 2)$. In this scenario, both players have a common least preferred chart but a different most preferred chart. In this case, the first player can deliberately eliminate their second least preferred chart, thereby forcing the second player to eliminate the first player's least preferred chart, on behalf of the first player. The payoff permutation $(a_2, b_2, c_2) = (1, 3, 2)$ is the only case when the first player to act eliminates a chart that is not their least preferred chart. In other payoff permutations, the first player to act always eliminates their least preferred chart. Such strategies are referred to as "sincere" strategies in the literature [21]. Notice that even in this case, it is guaranteed that the least preferred chart of either player is never chosen.

Additionally, the payoff $(a_2, b_2, c_2) = (1, 3, 2)$ is the only case where the player first to act has an advantage. In all other cases, the chart chosen is determined at the point when the payoffs are set, regardless of the order of the players' actions. Due to this fact, it can be concluded that when the Sequential Veto rule is used, it is desirable to determine the turn order probabilistically, in order to remove the advantage of the first action advantage.

Table 2
Payoff matrix for Players 1 and 2 in the Simultaneous Veto rule.

		Player 2		
		A	B	C
Player 1	A	$\frac{b_1+c_1}{2}, \frac{b_2+c_2}{2}$	c_1, c_2	b_1, b_2
	B	c_1, c_2	$\frac{a_1+c_1}{2}, \frac{a_2+c_2}{2}$	a_1, a_2
	C	b_1, b_2	a_1, a_2	$\frac{a_1+b_1}{2}, \frac{a_2+b_2}{2}$

5.4. Optimal strategies in the sequential veto rule

Summarizing the analysis the optimal strategy for the Sequential Veto rule is as follows:

- If the least preferred charts are the same for both players, but the most preferred charts are different for both players, the player first to act chooses the *second* least preferable chart. This forces the second player to act to choose their least preferable chart on behalf of the first player to act.
- In all other payoff designs, the sincere strategy is optimal — both players choose their least preferable charts remaining at that time.

Notice that there is one special case where the first player to act has an advantage. Therefore, in the eyes of the tournament designer, it becomes crucial that both players do not have a common least preferred chart when using the Sequential Veto rule, in order to remove the advantage of the player acting first. In all other cases, the optimal strategy for both players is always the “sincere strategy” of eliminating the least preferred chart for each player.

6. Analysis of simultaneous veto

In this section, we analyze the rules of the Simultaneous Veto rule, as described in Section 3.4.

6.1. Problem formulation

In the Simultaneous Veto rule, players simultaneously vote for one chart, similar to the Mind Game rule. Following the previous sections, we use the same settings and variables for the payoff function as in the Mind Game rule in Section 4.2.

In this rule, players simultaneously vote for a chart to be removed from a given chart pool, and the remaining chart is used for the match. If there are multiple charts remaining in the pool, the output chart is chosen randomly from the remaining charts with equal probability. For comparing with the other rules, we focus on a three-chart pool with two players, where each player votes once.

Based on the rules, the payoff matrix for this game can be modeled as Table 2. The matrix is mostly identical to that for the Mind Game rule shown in Section 4.1, with the exception of differences in the diagonal elements. The diagonal elements indicate cases when both players choose the same chart. Since the remaining charts are chosen at equal probabilities in these cases, the corresponding payoffs are set as the average payoffs of the remaining charts.

Following the previous sections, let us assume without loss of generality that $a_1 < b_1 < c_1$. Under this assumption, the strategy C for Player 1 is strictly dominated by other strategies, since $c_1 > b_1, c_1 > a_1$, and $b_1 > (a_1 + b_1)/2$. Therefore, the bottom row can in fact be removed to simplify the payoff matrix. For further analysis, we must assume concrete payoff values for Player 2.

Table 3
The payoff matrix for the classic game of Hawk–Dove (also known as Chicken).

		Player 2	
		Dove	Hawk
Player 1	Dove	3,3	1,4
	Hawk	4,1	0,0

Table 4

Payoff matrix for Players 1 and 2 in the Simultaneous Veto rule, when the payoff rankings are $(a, b, c) = (1, 2, 3)$ and $(r, s, t) = (1, 3, 2)$. In this case, dominated strategies can be removed from the original payoff matrix in Table 2, making this game isomorphic to the well-studied game of Hawk–Dove [22] (also known as Chicken).

		Player 2	
		A	C
Player 1	A	$\frac{b_1+c_1}{2}, \frac{b_2+c_2}{2}$	b_1, b_2
	B	c_1, c_2	a_1, a_2

6.2. Equilibrium analysis

Following the analysis of the Sequential Veto rule, assume that each player's chart payoffs are a permutation of $\{1, 2, 3\}$. Since we have assumed $a_1 < b_1 < c_1$ without loss of generality, there are six possible payoff configurations for Player 2. The Nash equilibria for each case is as follows:

- $(a_2, b_2, c_2) = (3, 2, 1), (3, 1, 2)$, and $(2, 3, 1)$ (least preferred chart differs, most preferred chart differs): There is a unique pure strategy Nash Equilibrium, where both players follow a sincere strategy.
- $(a_2, b_2, c_2) = (2, 1, 3)$ (least preferred chart differs, most preferred chart is the same): Two pure strategy Nash equilibria and one mixed strategy Nash equilibrium mixing over two strategies. Both pure strategy Nash equilibria simultaneously maximizes both players' payoffs. One of these equilibria can be attained by both players following a sincere strategy.
- $(a_2, b_2, c_2) = (1, 2, 3)$ (identical payoffs): Two pure strategy Nash equilibria and one mixed strategy Nash equilibrium mixing over two strategies. Both pure strategy Nash equilibria simultaneously maximizes both players' payoffs. However, they *cannot* be attained by following a sincere strategy, since both players must cooperate on eliminating the two suboptimal charts, requiring the players to communicate on which strategy to choose.
- $(a_2, b_2, c_2) = (1, 3, 2)$ (least preferred chart is the same, most preferred chart differs): This is isomorphic to the well-studied game of Hawk–Dove [22] (also known as Chicken). Two pure strategy Nash equilibria and one mixed strategy Nash equilibrium mixing over two strategies. There is *no* Nash equilibrium that simultaneously maximizes both players' payoffs.

The first two cases are identical to the Sequential Veto case, in the sense that the global optimum can be attained by a sincere strategy. In these cases, both the players' strategies and the final outcome are identical in both rules.

When $(a_2, b_2, c_2) = (1, 2, 3)$, the optimal equilibrium cannot be attained by a sincere strategy. This is due to the fact that the sincere strategy only eliminates one chart, since the payoffs for both players are exactly identical. The goal in this case is to eliminate payoffs 1 and 2, to obtain a payoff of 3. Therefore, the players must communicate and decide which of the players choose which strategy, and cooperate together to attain payoff 3.

Table 5

The relationship between payoff settings and game dynamics in each chart selection method. The payoff settings represent the order of the payoffs for each chart. Without loss of generality, the payoffs for Player 1 (P1)'s charts A, B, and C are set to 1, 2, and 3, respectively. In the Sequential Veto rule, only the row labeled "C, B" varies in outcome depending on the order of actions: if P1 goes first, chart C is chosen, and if Player 2 (P2) goes first, chart B is chosen. "HD" indicates that the game is isomorphic to the well-studied *Hawk-Dove* game (also known as *Chicken*), where there are multiple equilibria to choose from, imposing the equilibrium selection problem.

P1 payoffs (A, B, C)	P2 payoffs (A, B, C)	Mind Game Result	Sequential Veto			Simultaneous Veto			Notes
			Result	P1 Util.	P2 Util.	Result	P1 Util.	P2 Util.	
1, 2, 3	3, 2, 1	A, B, C	B	2	2	B	2	2	Zero Sum Game
1, 2, 3	2, 3, 1	A, B, C	B	2	3	B	2	3	
1, 2, 3	3, 1, 2	A, B, C	C	3	2	C	3	2	
1, 2, 3	1, 3, 2	A, B, C	C, B	3, 2	2, 3	HD	HD	HD	Common min. payoff charts
1, 2, 3	2, 1, 3	C	C	3	3	C	3	3	Common max. payoff charts
1, 2, 3	1, 2, 3	C	C	3	3	C	3	3	Common max. payoff charts

6.3. Relations with the game of Hawk-Dove

When $(a_2, b_2, c_2) = (1, 3, 2)$, this game is isomorphic to the well-studied game of *Hawk-Dove* [22] (also known as *Chicken*), shown in Table 3. The payoff matrix for Simultaneous Veto shown in Table 2 can simply be transformed into the matrix for Hawk-Dove by eliminating dominated strategies, which are C for Player 1 and B for Player 2. The payoff matrix then simplifies to Table 4. Since $a_1 < b_1 < (b_1 + c_1)/2 < c_1$ and $a_2 < c_2 < (b_2 + c_2)/2 < b_2$ holds by our assumptions, this game is isomorphic to the game of Hawk-Dove.

The game of Hawk-Dove is a two-player game, where each player has two strategies, Hawk and Dove. Hawk is an aggressive strategy and Dove is a gentle strategy. If one player acts aggressively and the other gently, then the aggressively acting player wins a larger payoff than the other "losing" player. However, if both players act aggressively, they both gain their least possible payoffs. When both players act gently, they gain a medium payoff, which is larger than the "losing" payoff when the strategies differ, but less than the winning payoff.

The analogy between Simultaneous Veto and Hawk-Dove can be found as follows. In Simultaneous Veto, since both players least prefer the same chart A, removing chart A (thus, strategy A) is the gentle strategy. On the other hand, removing the opponent's preferred chart is the aggressive strategy — if both players choose the aggressive strategy, the remaining chart is the least preferred chart, which yields the least possible payoffs. Here, this least preferred chart can be seen as a "catastrophic" chart, which both players want to avoid. The tension of the game then lies in whether to avoid this catastrophe by choosing Dove, or to take risks in maximizing your payoff by eliminating the opponent's preferred chart, in the hope that the opponent will eliminate the catastrophic chart in behalf of yourself.

As mentioned in the previous section, there is no Nash equilibrium that simultaneously maximizes both players' payoffs. This is different from the previous games, where there was an equilibrium that globally maximized both players' payoffs at once. Therefore, identifying a truly "optimal" strategy in this context becomes challenging. However, both players removing the least preferred chart (playing Dove) is a *maximin* strategy, which maximizes the worst possible outcome, or in other words minimizes their worst possible loss.

7. Comparisons in specific scenarios

Table 5 compares the game dynamics for each method, obtained from the analyses in the previous sections. For a more detailed comparison, here we will choose various concrete scenarios, and compare the dynamics of each rule in each setting. The scenarios we will compare are the following:

1. *Zero-sum setting*. Corresponds to (1, 2, 3), (3, 2, 1) in Table 5. We assume that both players use the win rate for each chart, with $a_1 = 1 - a_2$, $b_1 = 1 - b_2$, $c_1 = 1 - c_2$.
2. *Catastrophic Chart setting*. Corresponds to (1, 2, 3), (1, 3, 2) in Table 5. There is one "catastrophic chart" that globally minimizes both of the players' utilities, which both players want to avoid.

Table 6

The expected payoffs for the zero-sum setting. The utilities are $a_1 < b_1 < c_1$ for Player 1 and $a_2 > b_2 > c_2$ for Player 2.

Rule	P1 Payoffs	P2 Payoffs	Properties
Mind Game	$(a_1 + b_1 + c_1)/3$	$(a_2 + b_2 + c_2)/3$	Average
Sequential Veto	b_1	b_2	Median
Simultaneous Veto	b_1	b_2	Median

3. *Cooperative setting*. Corresponds to (1, 2, 3), (2, 1, 3) and (1, 2, 3), (1, 2, 3). There is one best chart that globally maximizes both of the players' utilities.

Note that the second "catastrophic chart" setting *never* occurs in the zero-sum case — in other words, in the zero-sum case, there cannot exist a unique chart that minimizes the utility for both players. This can simply be shown by contradiction. If we suppose that such a chart exists, the utilities of all charts must be equal for each player, due to the zero-sum setting. Since the chart does not become unique, this contradicts with our assumptions, thus making this case impossible in the zero-sum setting. A similar calculation shows that the third "cooperative setting" also never occurs in the zero-sum case. Therefore, these cases must be analyzed separately.

7.1. Zero-sum setting

This is the most straightforward setting — an example is where both players are aware of each others' performances for all charts, and both players maximize their win rates. Table 6 summarizes the expected payoffs in this case. In this case, Sequential Veto and Simultaneous Veto are functionally similar.

Notably, the expected payoffs for the Mind Game rule is the *average* of all charts, and for the Sequential and Simultaneous Veto rules are the *median* of all charts, for both players.

7.1.1. Skill gap fairness assessment for the veto rules

Let the "skill gap" $G(X)$ for a chart X be the absolute difference of the winning probabilities for each player. The skill gaps for charts A, B, and C are thus $G(A) = |a_1 - a_2|$, $G(B) = |b_1 - b_2|$, and $G(C) = |c_1 - c_2|$, respectively. One possible criterion for a fair match is minimizing the skill gap between players, thereby reducing the likelihood of a one-sided outcome. This also contributes in reducing the predictability of the outcome of the match, potentially increasing tension and suspense within the spectators and the players. This thus rather may be a criterion in the perspective of the spectators instead of the players, since the interest of the players are the raw winning probabilities.

At first sight, it may seem that the Veto rules minimize the skill gap between the players. However, this is in fact not necessarily the case. This stems from the fact that $\arg \min_X G(X)$ does not necessarily equal B. Some simple examples are $(a_1, b_1, c_1) = (0.1, 0.2, 0.4)$ and $(a_1, b_1, c_1) = (0.1, 0.2, 0.6)$, where in both cases $\arg \min_X G(X) = C$ with $\min_X G(X) = |0.6 - 0.4| = 0.2$. Since the Veto rules always choose chart B, this means

that the Veto rules do not necessarily choose the chart with the smallest skill gap.

Then, when do the Veto rules minimize the skill gaps? By writing the winning probabilities for chart X as x_1 and x_2 for Players 1 and 2, respectively,

$$G(X) := |x_1 - x_2| = |x_1 - (1 - x_1)| = |2x_1 - 1|, \quad (1)$$

since $x_1 = 1 - x_2$. Therefore,

$$\arg \min_X G(X) = \arg \min_X |2x_1(X) - 1| = \arg \min_X \left| x_1(X) - \frac{1}{2} \right|, \quad (2)$$

where we have written $x_1(X)$ to emphasize that x_1 depends on X . This means that charts minimizes the skill gap if and only if that chart has the winning probability closest to $1/2$. Since substituting $x_1 = 1 - x_2$ in Eq. (2) yields the same expression for x_2 , the proximity to $1/2$ holds for both players if it holds for either one.

This is particularly against the interests of a strictly losing player, which we define as a player whose winning probabilities for all charts are less than the opponent. In this case, $a_1 < b_1 < c_1 < 1/2$, so $\arg \min_X G(X) = C$, since c_1 is always the closest to $1/2$. Since chart C is always removed by the opponent, the Veto rule rather reinforces the already existing skill gap between the players rather than minimizing it.

For players that have a more closer skill set, it can be expected that the chances of a player is strictly losing against the opponent. Empirically, this should also increase the probability for the median winning probability to be closest to $1/2$, making the Veto rules fair in the skill gap criterion.

Therefore, in the perspective of the skill gap criterion, the Veto rule reinforces the skill gaps when used against a strictly losing player, and is expected to minimize the skill gap when used for players with close skill sets.

7.1.2. Optimal rule preferences

The average and median probabilities can both be larger than each other, depending on the probabilities. Therefore, given the winning probabilities, a player must have a preference for one rule over the other that favors their expected winning probabilities of the match. Here we analyze the preferences of the player.

A player prefers the Mind Game rule when the average winning probability is larger than the median probability. This holds when

$$\frac{a+b+c}{3} > b, \quad (3)$$

where we have written a, b, c for the probabilities, for simplicity. By simple arithmetic we find that this is equivalent to

$$c - b > b - a. \quad (4)$$

From $a < b < c$, both sides are positive, thus this is equivalent to

$$|c - b| > |b - a|. \quad (5)$$

Therefore, whether or not a player prefers the Mind Game rule is dictated by whether if b is closer to a or c . If b is closer to a , then the Mind Game rule is preferred. If b is closer to c , the Veto rules are preferred.

An example of a case when b is closer to a is a player that have very low winning probabilities for two charts, but a very high probability for one chart, such as $(a, b, c) = (0.1, 0.2, 0.9)$. We refer to this as a *polarized* player. Such players may appear when a player allocates most of their practice time into one chart, leaving the other charts unpracticed. Therefore, polarized players may be a model for certain types of newcomers, showing that the Mind Game rule is in favor of these types of newcomers.

Another model of a newcomer player is the strictly losing player. Interestingly, the strictly losing player does not necessarily prefer the Mind Game rule in terms of the expected utility, since b may not be closer to a than to c , even if $a < b < c < 1/2$.

Table 7

The expected payoffs for the Catastrophic Chart setting. The utilities are $a_1 < b_1 < c_1$ for Player 1 and $a_2 < b_2 < c_2$ for Player 2. In the Sequential Veto rule, we assume that Player 1 acts first in declaring a chart to be eliminated from the pool.

Rule	P1 payoffs	P2 payoffs
Mind Game (Nash)	$(a_1 + b_1 + c_1)/3$	$(a_2 + b_2 + c_2)/3$
Sequential Veto (Subgame Perfect)	c_1	c_2
Simultaneous Veto (Nash)	Hawk–Dove	Hawk–Dove
Simultaneous Veto (Mutual Maximin)	$(b_1 + c_1)/2$	$(b_2 + c_2)/2$

7.2. Catastrophic chart setting

The Catastrophic Chart setting is when both players agree on the chart that minimizes their payoffs (in this case, chart A), but do not agree on the chart that maximizes their payoffs (B and C). Such utility settings can occur when there is a *secondary* utility that comes into play, other than the winning probabilities. We hence refer to chart A as the “catastrophic” chart, since it carries a burden large enough to influence the utility of the players. Recall that this does not occur in the zero-sum case, since such a chart cannot be unique.

Here, assuming the perspective of the players who want to avoid such charts as much as possible, we assess the avoidability of the catastrophic chart within the given rules, using the expected payoffs stated in Table 7. Under this criterion, the most robust rule is Sequential Veto, followed by Simultaneous Veto, and lastly, Mind Game. In Sequential Veto, the subgame perfect equilibrium always avoids the catastrophic chart with a 100% probability. There is a unique *maximin* strategy in Simultaneous Veto, which eliminates the catastrophic chart, even though the rule do not have Nash equilibrium that globally maximizes the utility of all players. The most dangerous rule is the Mind Game rule, where the catastrophic chart is unavoidable in the Nash equilibrium. Regardless of how catastrophic the chart is, or in other words, for an arbitrary low value of a_1 and a_2 , as long as the maximum probability charts do not agree, the Nash equilibrium is to declare all charts with equal probabilities.

Note that a low utility for the players does not necessarily imply a low utility for the spectators as well. This case often arises if the reason for a catastrophic chart is the difficulty. Since such difficult charts require a high level of skill, the spectators may enjoy watching the match for the catastrophic charts. Therefore, if leaving a possibility for the catastrophic chart to be chosen is in particular interest, especially for the spectators, then the Mind Game rule may be the best choice among these rules, since it is the only rule among these that allow the choice of the catastrophic chart in the equilibrium solution.

7.3. Cooperative setting

The final case is when the order of the utility functions coincide, meaning that $a_1 < b_1 < c_1$ and $a_2 < b_2 < c_2$. In this case, the equilibrium strategy for all rules choose chart C with a 100% probability.

Note however that for the Mind Game and Simultaneous Veto rules, players must be able to communicate to decide which strategy to execute in order to reach the equilibrium. This is since for both rules, there are multiple pure strategy equilibria that realize the same maximum utility. This communication problem does not occur in the Sequential Veto rule, since the second acting player can choose their action conditionally based on the action of the first player.

8. Discussions

In this section, we compare and evaluate the fairness and strategic depth of each method. To this end, we first summarize the game dynamics under the optimal strategies for each selection method. Based on these insights, we discuss the scenarios where each rule is best suited for.

8.1. The optimal strategy for each method

The optimal strategy for each method can be summarized as follows:

Mind Game:

- If the payoff maximizing chart (let it be C) for both players is identical, both players declaring C is optimal. The output chart will always be chart C .
- In all other cases (including the zero-sum case), declaring each chart with equal probability is optimal. The output chart will be each of the three charts, each appearing with equal probabilities.

Sequential Veto:

- If the payoff minimizing chart for both players is identical, the first player to act declares the chart with the *second* highest payoff. The player to act second then chooses the chart with the lowest payoff.
- In all other cases (including the zero-sum case), it is optimal for both players to declare the chart that minimizes their own payoffs at that point, regardless to the order of action.

Simultaneous Veto:

- When the payoff minimizing chart for both players is identical but the payoff maximizing charts are different, the game becomes isomorphic to the game of *Chicken* (also known as *Hawk–Dove*). A definite optimal strategy is thus difficult to define in this case, but the sincere strategy is a maximin strategy, which maximizes the worst possible payoff.
- When the payoffs are identical for both players, the sincere strategy does not work — the players must cooperate in eliminating the two charts with lower payoffs. When doing so, the players must agree on which chart each of them will eliminate.
- In all other cases (including the zero-sum case), it is optimal for both players to use the sincere strategy, by declaring the chart that minimizes their own payoffs.

These optimal strategies stated in this paper could be a spoiler for the players. However, even with the optimal strategies, we find that the playabilities of those methods (e.g., estimating the opponent's preference, randomness, or suspense) still exist. This fact ensures that those methods can be used in future tournaments where the optimal strategies are widely known to every player.

8.2. Suitable tournament scenes

Comparing the dynamics of each rule, we can see that each game respects a different notion of fairness. Here we will discuss the suitable tournament scenes for each rule.

The Mind Game rule is fair in the sense that in the most usual zero-sum case, all charts can be chosen with an equal probability. This creates a sense of uncertainty, keeping the players in suspense until the final selection. Thus, if the goal for the tournament is to heighten suspense and maintain tension throughout the selection process, the Mind Game rule can be considered as a more effective choice.

As we have seen in Section 7.1.2, this is advantageous for the *polarized* player, who have two inferior charts and one proficient chart. This provides a model for a type of newcomers who have practiced most of their time on one chart but have not practiced the rest. Therefore, even for players with an inferior skill than the opponent, the chart most suitable for the disadvantageous player may be chosen in the Mind Game rule. This makes more room for newcomers to win the match, which may possibly bring excitement to the tournament.

On the other hand, the Sequential and Simultaneous Veto rules ensure fairness by guaranteeing that neither player's least preferred chart is ever chosen. As we have seen in Section 7.1.1, in the zero-sum setting, this is disadvantageous for a strictly losing player, who have

lower winning rates for all of the charts. Therefore, in the perspective of the players, the Veto rules may be considered fair when the skills of the players are more closer.

Another perspective in comparing the game dynamics is the interpretability of the players' actions, in the perspective of the spectators. In the Sequential and Simultaneous Veto rules, excluding rare cases, players generally follow a sincere strategy, where each player removes their least preferred charts. Since the actions depend on the players' preferences, the spectators of the match can infer each player's strengths, weaknesses, and style based on their declared chart choices. This allows room for the spectators to associate the players' choices with certain characters, making the viewing experience more engaging. In contrast, in the Mind Game rule, the choices of the players do not necessarily reflect their true preferences in the equilibrium strategy, except when both players share the same top choice. Therefore, in events where the interpretability of actions is considered valuable, the Veto rules are more suitable for the purpose.

8.3. Limitations and future work

8.3.1. Payoff design

One limitation of our analysis is that we assume that the payoffs for both players are common knowledge. This is not always the case, since the player skills and the opponent's belief over each other's skills are usually hidden knowledge, which impacts the payoff calculations.

8.3.2. Generalizing the chart pool size

Generalizing the number of charts is also a room for improvement in our study. In Sections 3–6, we fixed the number of charts as one for the match and three for the candidate list. However, this could be generalized into a larger chart pool. For example, the “LADDER MATCH” mode in the game *DJMAX RESPECT V*, which we mentioned in Section 3, uses the Simultaneous Veto rule, although there are five charts in the list, three of which are used for the match.

8.3.3. Considering effects between multiple matches

The current analysis only focuses on one match, and ignores the effect between multiple matches. Inter-match effects can be large especially in games such as *DDR*, where players use a lot of physical energy for each chart. In such settings, the allocation of physical energy and its trade-off between performance becomes important. For example, it may be possible for players to choose charts that use less physical energy, to save energy for later stages in the tournament to enhance their performance. Such effects can be considered as future courses of our research.

8.3.4. Application to other genres

All rules discussed in our paper share a simple structure where each player makes exactly one decision. These rules allow determining the final chart in a single step without requiring repeated actions, such as resolving ties in *Rock–Paper–Scissors*. Due to this simplicity, both selection methods fit well within the time constraints of tournaments, which is likely a key factor in their widespread adoption in the music game tournament scene.

Beyond rhythm games, the decision-making scenarios where players must select from a given set of options are roughly divided into two categories: “the selection of things that could be freely chosen by each player” and “the selection of shared environment between the players”. One example of the former scenario is the character selection in *MOBA* games, which often follows a sequential elimination variant known as *BAN & PICK* [11]. The chart selection in rhythm game tournaments exemplifies the latter scenario. Considering the nature of each rule, Mind Game is only applied to the latter scenario, while Sequential Veto and Simultaneous Veto might be used in both scenarios. This could be one reason that the Mind Game is hardly seen in the tournament other than the rhythm game.

As mentioned in Section 3, the only rule that confirmed the example of use in other game genres among the three discussed in this paper is Sequential Veto. Taking the example of the game VALORANT, the Sequential Veto is used for map selection. Regarding the shared environment among participants, the maps in VALORANT have a similar role to the charts in rhythm games. However, this process is only applied in the competitive scene and not in the casual playing scene,¹³ where the game system automatically decides the map. This implies that the rules for deciding a shared environment are essential in situations where fairness must be ensured. Furthermore, according to the discussion in Sections 7 and 8.2, Sequential Veto is more suitable for such situations than Mind Game; this might be one reason that we only find Sequential Veto in tournaments other than rhythm games.

Conversely, the Mind Game rule could be applicable in scenarios where participants must collectively decide on a single choice with a sense of uncertainty. One of the characteristics of rhythm game community tournaments is that the procedures with uncertainty are incorporated into shared environment decisions. Investigating the potential benefits of applications to the tournament of other genres remains an interesting subject for future research. To tackle this subject, it will be essential to collaborate with experts of the target game who have knowledge of (community) tournaments, like the second author of this paper who is familiar with the culture of rhythm games.

8.3.5. Benefits for players beyond the winning probabilities

Winning a match is not always the only goal of participating in a tournament. In this paper, we focus on the single match and assume the payoff for the player. The setting of payoffs is not limited to the winning probabilities, though it is a straightforward example. For instance, if the player senses what the audience wants, the player might relinquish the optimal action to win the match and meet the audience's expectations. The ambiguity in interpreting the payoffs allows the player to act in such a somewhat unreasonable way. However, the ambiguity makes the analysis difficult as it masks the player's payoffs. This difficulty might converge with the limitation mentioned in Section 8.3.1; therefore, it is a future work.

The benefit of participating in the tournament could exist even outside the match. For example, some players enjoy playing the game with famous players regardless of the result of the match. Talking with other players or audiences during the tournament's break time is also a good experience, especially for players participating from distant regions. These benefits arise from the fact that the tournament is based on the community of the players and cannot be expressed in terms of game-theoretic analysis of the match. In order to discuss the design of a tournament from such a macro perspective, it might be necessary to consider the emotional aspects of the players in addition to the game theory that we discussed in this paper.

9. Conclusion

In this paper, we analyzed the structure and strategic implications of chart selection methods in rhythm game tournaments. By mathematically modeling three widely used selection rules and applying game theory, we examined their fairness, strategic depth, and entertainment value.

This study makes several contributions. First, we documented and formalized tournament selection methods that have traditionally been shared informally within the community, providing a structured academic perspective on their design. Second, we analyzed the strategic characteristics of each rule, highlighting how different selection methods influence fairness and player interaction, thereby shaping the

overall tournament experience. Finally, we have discussed the potential for extending these rules and adapting similar selection mechanisms to competitive formats beyond rhythm games, such as other e-sports tournaments with structured decision-making elements.

Our findings suggest that different chart selection methods offer distinct advantages depending on the tournament's objectives—whether prioritizing fairness, maintaining suspense, or emphasizing player expression. The Mind Game rule enhances unpredictability and tension, while the Sequential and Simultaneous Veto rules ensure a level of fairness by eliminating least-preferred options. Understanding these dynamics provides valuable insights for tournament organizers seeking to balance competition and entertainment.

Future research could explore variations of these selection methods, especially in settings where players lack complete information about each other's preferences or where negotiation and deception play a role in the decision-making process.

CRedit authorship contribution statement

Hikaru Ikuta: Writing – original draft, Validation, Methodology, Formal analysis. **Yudai Tsujino:** Writing – original draft, Investigation, Data curation, Conceptualization. **Ryosuke Yamanishi:** Writing – review & editing, Supervision.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT, DeepL, and Grammarly in order to improve the readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors have declared no conflict of interest.

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Appendix. Enumeration of nash equilibria in the mind game using support enumeration

In two-player matrix-form strategic games, if the game is *nondegenerate*, an algorithm called support enumeration can be used to list all Nash equilibria [17]. A nondegenerate game is one where any strategy with a support size of k has no more than k pure strategies as best responses. The support of a strategy refers to the set of pure strategies executed with non-zero probability.

This section explains how to enumerate all Nash equilibria of the Mind Game using the support enumeration algorithm. For brevity, only an outline of the proof is provided. The proof consists of two steps: first, we establish that the game is nondegenerate, and then we enumerate the Nash equilibria. In nondegenerate games, the support sizes of the players' strategies in Nash equilibrium are always equal [17], so it suffices to analyze cases where the support sizes are 1, 2, and 3.

As previously stated, this paper assumes that the payoff values for Player 1 satisfy $0 \leq a_1 < b_1 < c_1 \leq 1$, while the payoffs a_2, b_2, c_2 for Player 2 are distinct and satisfy $0 \leq a_2, b_2, c_2 \leq 1$. For simplicity, Player 1 is referred to as “self”, and Player 2 as “opponent”, and will write $a := a_1, b := b_1, c := c_1$.

A.1. Proof of nondegeneracy of the game

To prove that the game is nondegenerate, we must show that any strategy with support size k has no more than k pure strategies as best responses. Since the game has three strategies, we only need to consider $k = 1$ and $k = 2$. Specifically, for $k = 2$, we must demonstrate that the number of best responses cannot be 3.

¹³ VALORANT Systems Health Series - Map Diversity <https://playvalorant.com/en-us/news/dev/valorant-systems-health-series-map-diversity/> <Retrieved on Apr. 28, 2025>.

Case $k = 1$:

Proof. In the Mind Game, for any pure strategy of Player 1, there exists a pure strategy of Player 2 that derives each of the charts A, B, and C. Since the payoffs for each chart a_2, b_2, c_2 are distinct, the opponent's best response is uniquely determined, ensuring that more than one best response cannot occur. This reasoning holds symmetrically if Player 1 and 2 are interchanged. Thus, the case $k = 1$ is proven. $\square$

Case $k = 2$:

Proof. We prove this by contradiction. Assume Player 2's strategy has a support size of 2, excluding strategy A. Further, assume that Player 1 has 3 best responses. Since the payoffs for these best responses are equal, the following equation holds:

$$\begin{pmatrix} c-b & b-a \\ c-a & b-c \end{pmatrix} \begin{pmatrix} q_B \\ q_C \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (\text{A.1})$$

If the coefficient matrix on the left-hand side is invertible, $q_B = q_C = 0$, which is invalid because it does not form a valid probability distribution. Hence, the matrix must be singular, leading to:

$$\begin{vmatrix} c-b & b-a \\ c-a & b-c \end{vmatrix} = -a^2 - b^2 - c^2 + ab + bc + ca = 0. \quad (\text{A.2})$$

Considering this as a quadratic equation in a , the discriminant Δ_a is:

$$\Delta_a = -3b^2 - 3c^2 + 6bc = -3(b-c)^2.$$

Since $a \in \mathbb{R}$, $\Delta_a \geq 0$ must hold. However, $\Delta_a = 0$ requires $b = c$, which when substituted into Eq. (A.2), implies $a = b = c$. This contradicts the assumption that a, b, c are distinct.

Thus, by contradiction, when the opponent's strategy excludes A and has a support size of 2, "self" cannot have 3 best responses. Similar contradictions arise when B or C are excluded, confirming that no strategy of support size 2 can have 3 best responses.

Since the game's payoff matrix is symmetric, the same reasoning applies when Players 1 and 2 are interchanged. Thus, the case $k = 2$ is proven. $\square$

With both cases $k = 1$ and $k = 2$ established, it follows that under the assumption of distinct payoffs, the Mind Game is nondegenerate.

A.2. When both players' strategies have support of 1

This corresponds to pure-strategy Nash equilibria. As shown earlier, for any pure strategy of Player 1, there is a corresponding pure strategy of the opponent for each payoff r, s, t . If the most preferred songs of both players do not align, no pure-strategy Nash equilibrium exists. Conversely, if their most preferred songs align, three Nash equilibria emerge, corresponding to the strategy pairs (A, B), (B, A), (C, C), where the most preferred song is selected.

A.3. When both players' strategies have support of 2

Consider the case where both players exclude strategy C. If the strategy pair forms a Nash equilibrium, the expected payoffs for all pure strategies of Player 1 must be equal, leading to:

$$\begin{pmatrix} a-c & c-b \\ 1 & 1 \end{pmatrix} \begin{pmatrix} q_A \\ q_B \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Solving this yields:

$$\begin{pmatrix} q_A \\ q_B \end{pmatrix} = \frac{1}{2c-a-b} \begin{pmatrix} c-b \\ c-a \end{pmatrix}.$$

From this we obtain the condition $q_A, q_B > 0$. Similar derivations for other cases (excluding strategies A, B, C) show that Nash equilibria with support size 2 only occur when both players' most preferred song is C.

A.4. When both players' strategies have support of 3

In this case, the only Nash equilibrium is when both players assign equal probability to each strategy. This equilibrium exists regardless of the relative ordering of a_2, b_2, c_2 .

Data availability

No data was used for the research described in the article.

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